

Cloud LIMS — Supabase setup

Connect the pipeline to the cloud: create the database, wire the hub, and give testers a web app to create registrations. Companion to LAN-SETUP.md and LIMS-Integration-Guide.md.

Where things run (read first)

Your 60 client PCs have NO internet - testers do NOT use them for the LIMS. The web app + Supabase need internet, so testers create registrations on an internet-connected machine (their own PC, or the central desktop). Those registrations flow DOWN to the offline clients through the hub. Cloud = where registrations are born; hub = the bridge; clients = consumers. Adding Supabase flips the hub from LOCAL mode into CLOUD mode: it syncs the catalog FROM Supabase and forwards every filed PDF TO it.

1

Create the project

At **supabase.com** create a new project (pick a nearby region, set a DB password). Then **Settings** → **API** and copy three values:

Value	Used by
Project URL (https://xxxx.supabase.co)	hub + web app
anon key (public)	web app ONLY
service_role key (secret)	hub ONLY

2

Create the database

SQL Editor → **New query** → paste all of `lims/supabase/schema.sql` → **Run**. It creates the tables (device_types, registrations, registration_tests, documents), row-level security, realtime, the private **lims-docs** storage bucket, and the `catalog_version()` watermark. Idempotent - safe to re-run.

3

Add your device types

The schema seeds only `gcms`, `lcms`, `icpms`. Add EVERY device type your 60 clients were enrolled with, or prints from those devices are held (device_type_mismatch). In the SQL Editor:

```
insert into public.device_types (id) values ('dsgd'), ('xrf')
on conflict (id) do nothing;
```

Find the exact type ids in the hub dashboard → **Catalog / Devices** tab.

4

Create tester logins

Authentication → **Providers** → enable **Email**. Then **Authentication** → **Users** → **Add user** for each tester (email + password). These are who sign into the web app.

5

Point the HUB at Supabase (this switches it to cloud mode)

On the central desktop open `http://localhost:8000/` → **Settings**: set **Supabase URL** = the Project URL, **Supabase service role key** = the service_role key (stored DPAPI-encrypted here, never leaves this machine) → **Save**. Within seconds Settings shows `supabase_configured` and the Catalog tab mirrors Supabase.

Two warnings

(1) Cloud sync **OVERWRITES** the hub's local catalog - any registrations you added on the hub in local mode are replaced by Supabase, so recreate them in the web app (Step 7). (2) The service_role key goes **HERE** and only here - never in the web app or git.

6

Deploy the tester web app

On any internet-connected machine:

```
cd lims\web
copy .env.example .env :: then edit .env:
VITE_SUPABASE_URL=https://xxxx.supabase.co
VITE_SUPABASE_ANON_KEY= :: anon ONLY, never the service key
npm install
npm run dev :: http://localhost:5173 (or: npm run build -> host dist\)
```

7

Testers create registrations

In the web app: **sign in** → **Registrations** → enter the registration number, pick the **device type**, add the **tests** as chips → Save. It reaches the hub and every client dropdown within about 2 seconds.

8

Verify the full loop

Check	Expect
Create reg in web app (e.g. R-2026-1000 / gcms / assay)	appears in the client dropdown in ~2 s
Client prints → pick reg + test → Attach & print	hub Documents: filed → forwarded
Web app → Documents page	the document appears (realtime); Download works
Supabase → Storage → lims-docs	the PDF object is there

Which secret goes where

Secret	Goes into	Never into
anon key	web app .env	the hub
service_role key	hub → Settings tab	the web app / git / any client

Common gotchas

Symptom	Cause & fix
Prints get held: device_type_mismatch	The registration's device type must exist in device_types AND match the printer's enrolled type - add it (Step 3).
Web app can't sign in	Email auth not enabled, or wrong Project URL / anon key in .env.
Documents don't reach the web app	Check the hub Documents tab for forward_failed (internet/keys) - it retries automatically. Confirm Settings shows supabase_configured.
Testers on the 60 clients can't open the web app	Expected - those PCs are offline. Use an internet-connected machine.
Old hub registrations vanished	Expected - cloud sync replaced the local catalog. Recreate them in the web app.
Catalog tab now read-only	Expected in cloud mode - manage registrations in the web app; the hub mirrors them.

Order of operations

Do Steps 1-5 first - that alone starts documents flowing to the cloud. Then Steps 6-7 give testers the web UI. No Supabase yet? The hub happily runs in LOCAL mode and you can switch to cloud any time; the forward queue drains the backlog automatically once configured.

Full request/response contracts: documents/LIMS-Integration-Guide.md (sections A & B). Schema: lims/supabase/schema.sql. Web app: lims/README.md.